

# EQ Bank Mobile Banking Application

## Architecture Document

### Components:

- Mobile App (iOS/Android) - customer-facing banking interface
- API Gateway - routes requests, rate limiting, authentication
- Authentication Service - OAuth2/OIDC, MFA, session management
- Account Service - balance inquiries, account details, statements
- Payment Service - e-Transfer, bill payments, internal transfers
- Database - PostgreSQL, stores account data, transaction history
- Message Queue - async processing for payments and notifications
- Notification Service - push notifications, email, SMS alerts

### Data Flows:

- Mobile App -> API Gateway: HTTPS API calls (REST)
- API Gateway -> Authentication Service: token validation
- API Gateway -> Account Service: account queries
- API Gateway -> Payment Service: payment requests
- Account Service -> Database: read account data
- Payment Service -> Database: write transaction records
- Payment Service -> Message Queue: async payment events
- Message Queue -> Notification Service: payment confirmations

### Trust Boundaries:

- Internet boundary: Mobile App <-> API Gateway
- Internal boundary: API Gateway <-> Backend Services
- Data boundary: Backend Services <-> Database